Tutorial 7

Question 1

Four Master's students — **Rohit, Aisha, Kalp, and Yogesh** — participate in a university reading club. Each student rates four selected books on a scale from 1 to 5, where a higher score indicates a stronger preference for the book.

The four books are:

1. *The Feynman Lectures on Physics — Richard Feynman*
2. *The Boy Who Loved — Durjoy Datta*
3. *Pride and Prejudice — Jane Austen*
4. *A Brief History of Time — Stephen Hawking*

Their preference scores are represented by the following rating matrix A:

Book / Student	**Rohit**	**Aisha**	**Kalp**	**Yogesh**
Feynman Lectures	4	3	5	2
The Boy Who Loved	2	5	3	4
Pride and Prejudice	3	4	2	5
A Brief History of Time	5	2	4	3

$$A = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 2 & 5 & 3 & 4 \\ 3 & 4 & 2 & 5 \\ 5 & 2 & 4 & 3 \end{bmatrix}$$

The columns of A correspond, in order, to Rohit, Aisha, Kalp, and Yogesh, while the rows correspond, in order, to the four books listed above.

To combine the students' opinions, the reading club assigns the following importance weights:

Student	**Weight**
Rohit	2
Aisha	1
Kalp	3
Yogesh	2

$$w = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix}$$

The overall weighted score of the four books is obtained by the matrix-vector product

$$s = Aw$$

where each entry of s represents the overall weighted score of the corresponding book.

Answer the following:

(a) State the dimensions of matrix A and vector w.

(b) Determine whether the product Aw is mathematically defined. Justify your answer using the dimensions of A and w.

(c) Calculate the matrix-vector product $s = Aw$ and express the result as a single column vector with four entries.

(d) Using the calculated weighted scores, identify the book or books having the highest overall weighted score.

Question 2: Spanning and Orthogonality

Consider the following two vectors in the three-dimensional vector space $\mathbb{R}^3$:

$$v_1 = (1, 2, -1), \qquad v_2 = (2, -1, 0)$$

Answer the following:

(a) Can the two vectors v_1 and v_2 span the entire vector space $\mathbb{R}^3$? Give a theoretical justification for your answer based on the number of linearly independent vectors required to span a three-dimensional vector space. No row reduction or detailed calculation is required.

(b) Determine whether the vectors v_1 and v_2 are orthogonal. Calculate the dot product $v_1 \cdot v_2$ and use the result to justify your conclusion.

Question 3: Rotation Matrix

Consider the following two-dimensional rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Let $\theta = 90^\circ$ and consider the vector

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Answer the following:

(a) Substitute $\theta = 90^\circ$ into the rotation matrix and write the resulting numerical matrix R.

(b) Use the rotation matrix to calculate the transformed vector $v' = Rv$ and write the result as a column vector.

(c) Plot or indicate the positions of v and v' in the xy-plane, and state the direction through which the vector has been rotated.

(d) Calculate the transpose $R^\top$ and verify whether $R^\top R = I$, where I is the 2×2 identity matrix.

(e) Based on your calculation in part (d), state whether the rotation matrix R is an orthogonal matrix. Justify your answer.

(f) Compare the lengths of v and v'. State what property of a rotation matrix this illustrates.

Question 4

A point $P(2, 3)$ is first rotated by 45° anticlockwise and then scaled using

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Find the final coordinates using matrix multiplication.

Question 5

A point $P(1, 2)$ undergoes the following transformations in order:

1. x-direction shear with $k = 2$
2. 90° anticlockwise rotation
3. Scaling by 2 in both directions

Find the **combined transformation matrix** and the final coordinates of P.